

Chemical Thermodynamics

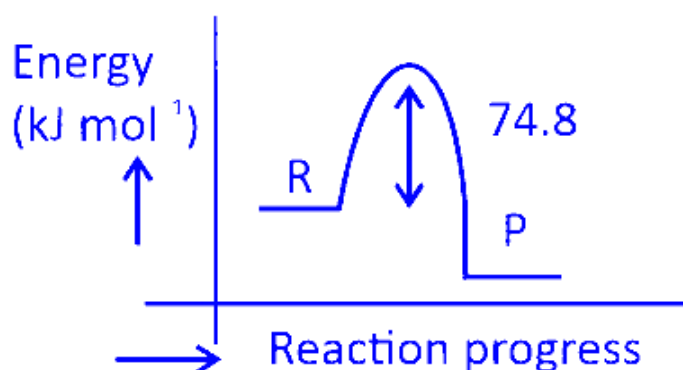
Question1

$\text{C(s)} + 2\text{H}_2(\text{g}) \rightarrow \text{CH}_4(\text{g}); \Delta H = -74.8 \text{ kJ mol}^{-1}$. Which of the following diagrams gives an accurate representation of the above reaction? [R $\rightarrow$ reactants; P $\rightarrow$ products]

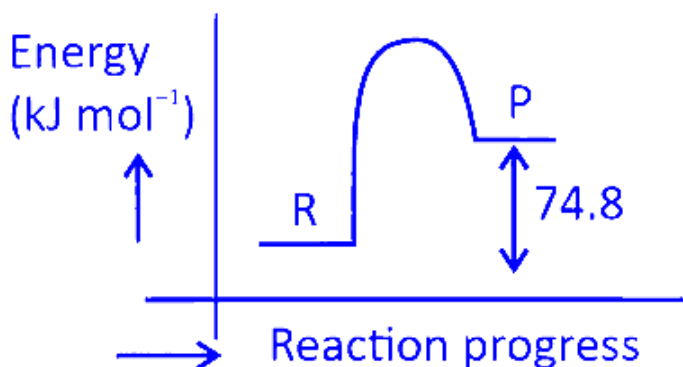
NEET 2025

Options:

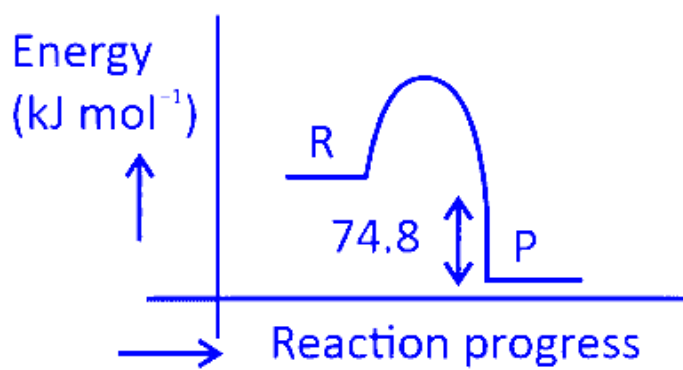
A.



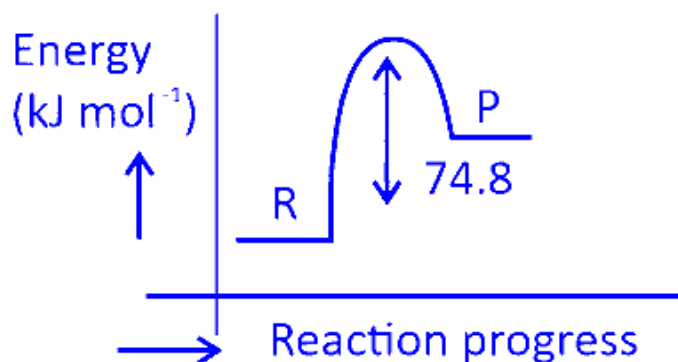
B.



C.



D.

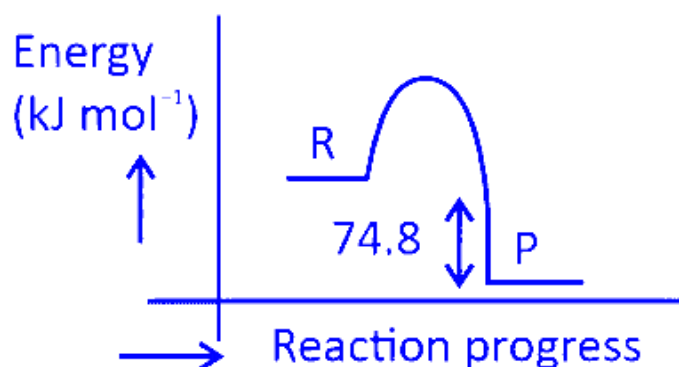


Answer: C

Solution:

$\Delta H = -74.8 \text{ kJ mol}^{-1}$, it is an exothermic reaction.

So, accurate representation is



Question2

The standard heat of formation, in kcal/mol of Ba^{2+} is : [Given : standard heat of formation of SO_4^{2-} ion (aq) = -216kcal/mol , standard heat of crystallisation of $\text{BaSO}_4(\text{s}) = -4.5\text{kcal/mol}$, standard heat of formation of $\text{BaSO}_4(\text{s}) = -349\text{kcal/mol}$]

NEET 2025

Options:

A. $+133.0$

B. $+220.5$

C. -128.5

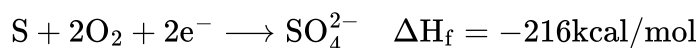
D. -133.0

Answer: C

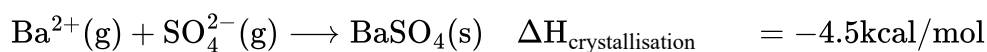
Solution:

Let's consider the following equations:

Formation of SO_4^{2-} :



Crystallization of $\text{BaSO}_4(\text{s})$:



Formation of $\text{BaSO}_4(\text{s})$:



We need to find the standard heat of formation for Ba^{2+} .

To determine this, apply the following relationship derived from the equations:



Using the data provided:

Start from equation (3), then subtract equations (1) and (2):

$$-349 - (-4.5) - (-216)$$

Simplifying the expression:

$$\begin{aligned}
 & -349 + 4.5 + 216 \\
 & = -349 + 220.5 \\
 & = -128.5 \text{ kcal/mol}
 \end{aligned}$$

This calculation shows that the standard heat of formation for Ba^{2+} is -128.5 kcal/mol .

Question3

Choose the correct statement for the work done in the expansion and heat absorbed or released when 5 litres of an ideal gas at 10 atmospheric pressure isothermally expands into vacuum until volume is 15 litres :

[NEET 2024 Re]

Options:

A.

Both the heat and work done will be greater than zero

B.

Heat absorbed will be less than zero and work done will be positive

C.

Work done will be zero and heat will also be zero

D.

Work done will be greater than zero and heat will remain zero

Answer: C

Solution:

Since it is isothermal, $\Delta T = 0$

$$\Delta U = nC_v \Delta T = 0$$

Since expansion is taking place against vacuum

$$P_{\text{ext}} = 0$$

$$W = -P_{\text{ext}} \Delta V = 0$$

From first law of thermodynamics,

$$\Delta U = q + W$$

$$0 = q + 0$$

$$q = 0$$

Question4

For an endothermic reaction:

- (A) q_p is negative.**
- (B) $\Delta_r H$ is positive.**
- (C) $\Delta_r H$ is negative.**
- (D) q_p is positive.**

Choose the correct answer from the options given below:

[NEET 2024 Re]

Options:

A.

B and D

B.

C and D

C.

A and B

D.

A and C

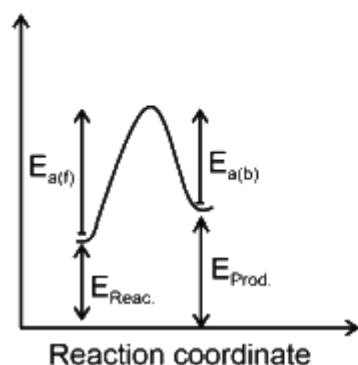
Answer: A

Solution:

For endothermic reactions,

$$\Delta H_r = \text{positive (Heat is absorbed)}$$

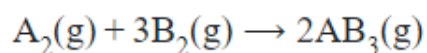
$$\Delta H = E_{a(f)} - E_{a(b)}$$



$$q_p = +ve$$

Question5

For the following reaction at 300 K



the enthalpy change is +15 kJ , then the internal energy change is :

[NEET 2024 Re]

Options:

A.

19988.4 J

B.

200 J

C.

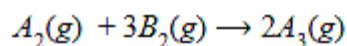
1999 J

D.

1.9988 kJ

Answer: A

Solution:



$$\Delta n_{(g)} = n_{(P)} - n_{(R)}$$

$$= 2 - 3 - 1 = -2$$

$$\Delta H = \Delta U + \Delta n_g RT$$

$$15 \times 1000 = \Delta U - 2 \times 8.314 \times 300$$

$$\Delta U = 15000 + 600 \times 8.314$$

$$= 15000 + 6 \times 831.4$$

$$= 15000 + 4988.4$$

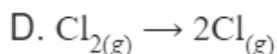
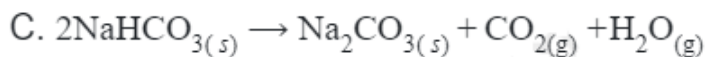
$$\Delta U = 19988.4 \text{ J}$$

Question6

In which of the following processes entropy increases?

A. A liquid evaporates to vapour.

B. Temperature of a crystalline solid lowered from 130K to 0K .



Choose the correct answer from the options given below:

[NEET 2024]

Options:

A.

A and C

B.

A, B and D

C.

A,C and D

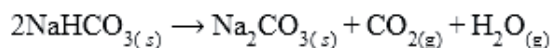
D.

C and D

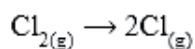
Answer: C

Solution:

When a liquid evaporates to vapour entropy increases.



Number of gaseous product molecules increases so entropy increases.



1 mole $\text{Cl}_{2(g)}$ form 2 mol $\text{Cl}_{(g)}$. So entropy increases.

Question7

Match List I with List II.

	List-I (Process)		List-II (Conditions)
A.	Isothermal process	I.	No heat exchange
B.	Isochoric process	II.	Carried out at constant temperature
C.	Isobaric process	III.	Carried out at constant volume
D.	Adiabatic process	IV.	Carried out at constant pressure

Choose the correct answer from the options given below:

[NEET 2024]

Options:

A.

A-IV, B-III, C-II, D-I

B.

A-IV, B-II, C-III, D-I

C.

A-I, B-II, C-III, D-IV

D.

A-II, B-III, C-IV, D-I

Answer: D

Solution:

(A) Isothermal process $\Rightarrow$ Temperature is constant throughout the process

(B) Isochoric process $\Rightarrow$ Volume is constant throughout the process

(C) Isobaric process $\Rightarrow$ Pressure is constant throughout the process

(D) Adiabatic process $\Rightarrow$ No exchange of heat (q) between system and surrounding

Question8

The work done during reversible isothermal expansion of one mole of hydrogen gas at 25°C from pressure of 20 atmosphere to 10 atmosphere is

(Given $R = 2.0 \text{ cal K}^{-1} \text{ mol}^{-1}$)

[NEET 2024]

Options:

A.

0 calorie

B.

-413.14 calories

C.

413.14 calories

D.

100 calories

Answer: B

Solution:

$$\begin{aligned}W_{\text{rev, iso}} &= -2.303nRT \log \frac{P_1}{P_f} \\&= -2.303 \times 1 \times 2 \times 298 \times \log 2 \\&= -2.303 \times 1 \times 2 \times 298 \times 0.3 \\&= -413.14 \text{ calories}\end{aligned}$$

Question9

Which amongst the following options is the correct relation between change in enthalpy and change in internal energy?

[NEET 2023]

Options:

A.

$$\Delta H = \Delta U + \Delta n_g RT$$

B.

$$\Delta H - \Delta U = - \Delta n RT$$

C.

$$\Delta H + \Delta U = \Delta n R$$

D.

$$\Delta H = \Delta U - \Delta n_g RT$$

Answer: A

Solution:

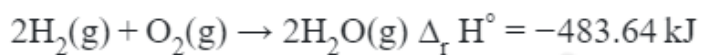
Solution:

Correct relation between change in enthalpy and change in internal energy is

$$\Delta H = \Delta U + \Delta n_g RT$$

Question 10

Consider the following reaction :-



What is the enthalpy change for decomposition of one mole of water? (Choose the right option).

[NEET 2023 mpr]

Options:

A.

120.9 kJ

B.

241.82 kJ

C.

18 kJ

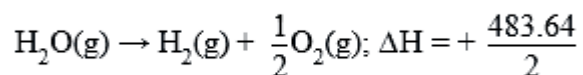
D.

100 kJ

Answer: B

Solution:

Decomposition for 1 mole of water



$$\Delta H = +241.82 \text{ kJ}$$

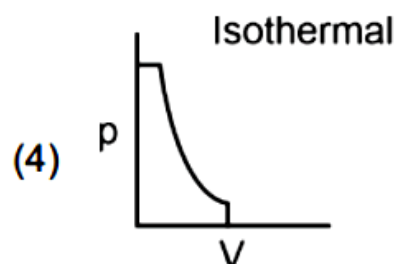
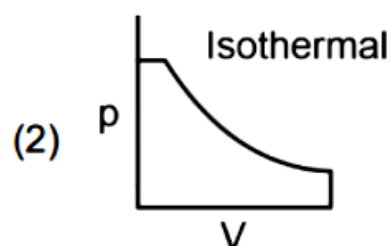
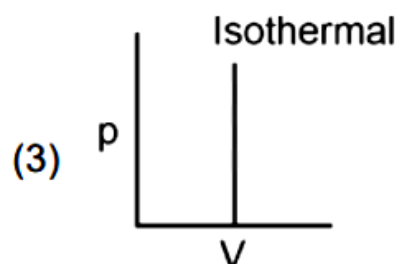
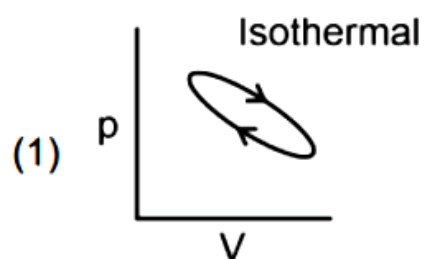
Question11

Which of the following $p - V$ curve represents maximum work done?

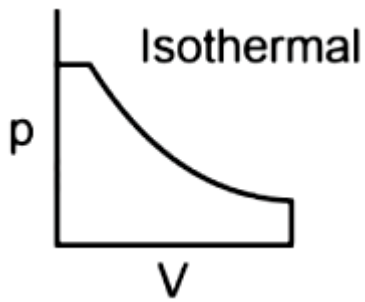
[NEET-2022]

Options:

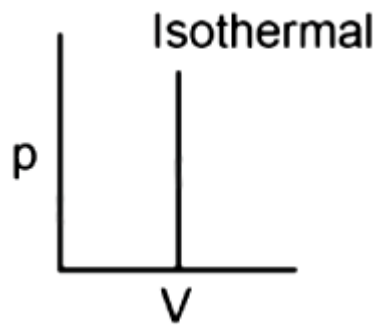
A.



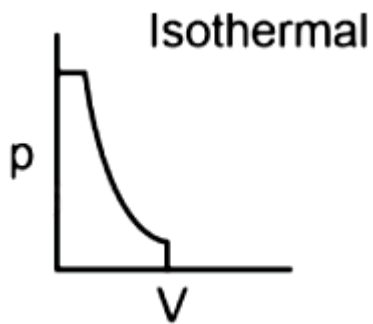
B.



C.



D.

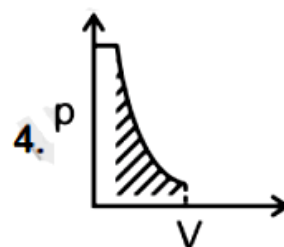
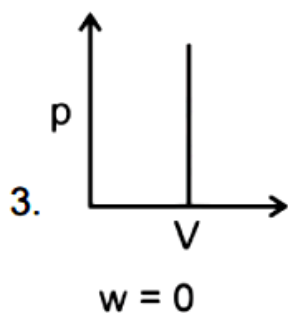
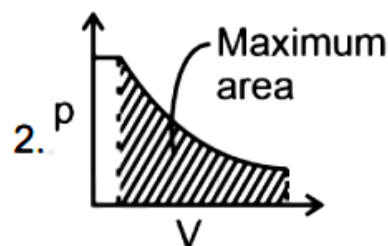
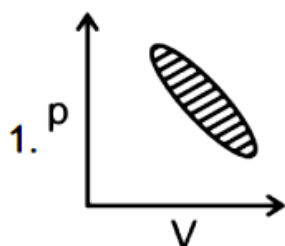


Answer: B

Solution:

Work done under any thermodynamic process can be determined by area under the 'p-V' graph.

As it can be observed maximum area is covered in option ' 2 '



Question12

One mole of an ideal gas at 300K is expanded isothermally from 1L to 10L volume. ΔU for this process is:

(Use $R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$)

[NEET Re-2022]

Options:

A. 0J

B. 1260J

C. 2520J

D. 5040J

Answer: A

Solution:

Solution:

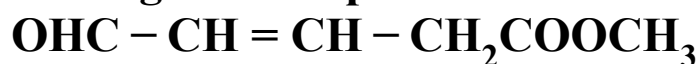
In isothermal expansion/compression of ideal gas

$$\Delta U = nC_v \Delta T, (\Delta T = 0)$$

$$\therefore \Delta U = 0$$

Question13

What is the hybridization shown by C_1 and C_2 carbons, respectively in the given compound?



[NEET Re-2022]

Options:

A. sp^3 and sp^3

B. sp^2 and sp^3

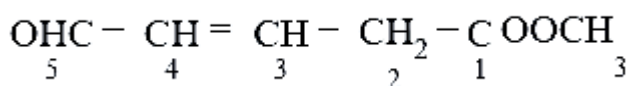
C. sp^2 and sp^2

D. sp^3 and sp^2

Answer: B

Solution:

Solution:



Ester group has more priority than aldehyde. So numbering should be done from left to right. C_1 has double bond and is sp^2 hybridised.

Question14

Which one among the following is the correct option for right relationship between C_p and C_v for one mole of ideal gas?

[NEET 2021]

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Options:

A. $C_p + C_v = R$

B. $C_p - C_v = R$

C. $C_p = RC_v$

D. $C_v = RC_p$

Answer: B

Solution:

Solution:

At constant volume, $q_v = C_v \Delta T = \Delta U$

At constant pressure, $q_p = C_p \Delta T = \Delta H$

For a mole of an ideal gas,

$$\Delta H = \Delta U + \Delta(PV)$$

On putting the values of ΔH and ΔU , we have

$$C_p \Delta T = C_v \Delta T + R \Delta T$$

$$C_p = C_v + R$$

$$C_p - C_v = R$$

Question15

For irreversible expansion of an ideal gas under isothermal condition, the correct option is:

[NEET 2021]

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Options:

A. $\Delta U = 0, \Delta S_{\text{total}} = 0$

B. $\Delta U \neq 0, \Delta S_{\text{total}} \neq 0$

C. $\Delta U = 0, \Delta S_{\text{total}} \neq 0$

D. $\Delta U \neq 0, \Delta S_{\text{total}} = 0$

Answer: C

Solution:

Solution:

- For a spontaneous process, $\Delta S_{\text{total}} > 0$ and since irreversible process is always spontaneous therefore $\Delta S_{\text{total}} > 0$.
 - Since $\Delta U = nC_v \Delta T$ and $\Delta T = 0$ for isothermal process therefore $\Delta U = 0$.
-

Question16

The correct option for free expansion of an ideal gas under adiabatic condition is
[2020]

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Options:

A. $q = 0, \Delta T < 0$ and $w > 0$

B. $q < 0, \Delta T = 0$ and $w = 0$

C. $q > 0, \Delta T > 0$ and $w > 0$

D. $q = 0, \Delta T = 0$ and $w = 0$

Answer: D

Solution:

Solution:

(d) Free expansion of ideal gas $P_{\text{ex}} = 0$

$$\therefore w = -P_{\text{ex}} \Delta V = 0$$

$\therefore$ Adiabatic process $\Rightarrow q = 0$

$\Delta E = q + w$ (first law of thermodynamics)

$$\therefore \Delta E = 0$$

$$\Delta E = nC_v dT \Rightarrow \Delta E = 0$$

$$\text{So, } q = 0, \Delta T = 0, w = 0$$

Question17

For the reaction, $2 \text{Cl(g)} \rightarrow \text{Cl}_2\text{(g)}$, the correct option is :
(2020)

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Options:

A. $\Delta_r H > 0$ and $\Delta_r S < 0$

B. $\Delta_r H < 0$ and $\Delta_r S > 0$

C. $\Delta_r H < 0$ and $\Delta_r S < 0$

D. $\Delta_r H > 0$ and $\Delta_r S > 0$

Answer: C

Solution:

Solution:

(c) We know that, $\text{Cl}_2\text{(g)} \rightarrow 2\text{Cl(g)}$ is endothermic reaction because it required energy to break bond.

So reverse reaction, $2\text{Cl(g)} \rightarrow \text{Cl}_2\text{(g)}$ will be exothermic, $\Delta_r H < 0$ Also, two gaseous atom combine together to form 1 gaseous molecule.

Question18

Under isothermal conditions, a gas at 300 K expands from 0.1 L to 0.25 L against a constant external pressure of 2 bar. The work done by the gas is [Given that 1L bar = 100J]
(NEET 2019)

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Options:

- A. 30J
- B. -30J
- C. 5kJ
- D. 25J

Answer: B

Solution:

Solution:

Expansion of a gas against a constant external pressure is an irreversible process. The work done in an irreversible process

$$\begin{aligned} &= -P_{\text{ext}} \Delta V = -P_{\text{ext}} (V_2 - V_1) = -2(0.25 - 0.1) \\ &= -2 \times 0.15 \text{ L bar} = -0.30 \times 100 \text{ J} = -30 \text{ J} \end{aligned}$$

Question19

**In which case change in entropy is negative?
(NEET 2019)**

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Options:

- A. $2\text{H}_{(\text{g})} \rightarrow \text{H}_{2(\text{g})}$
- B. Evaporation of water
- C. Expansion of a gas at constant temperature
- D. Sublimation of solid to gas

Answer: A

Solution:

Solution:

If $\Delta n_{\text{g}} < 0$ then $\Delta S < 0$

Question20

An ideal gas expands isothermally from 10^{-3}m^3 to 10^{-2}m^3 at 300K against a constant pressure of 10^5N m^{-2} . The work done on the gas is (Odisha NEET 2019)

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Options:

A. $+270\text{kJ}$

B. -900J

C. $+900\text{ kJ}$

D. -900kJ

Answer: B

Solution:

Solution:

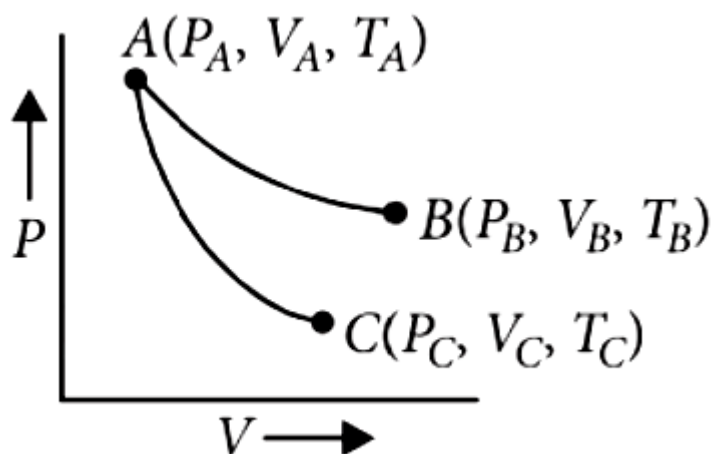
$$\begin{aligned}w &= -PdV = -P(V_2 - V_1) \\&= -10^5\text{N m}^{-2}(10^{-2} - 10^{-3})\text{m}^3 \\&= -10^5\text{N m}^{-2}(9 \times 10^{-3})\text{m}^3 \\&= -9 \times 10^2\text{N m} = -900\text{J} \quad (\because 1\text{J} = 1\text{N m})\end{aligned}$$

Question21

Reversible expansion of an ideal gas under isothermal and adiabatic conditions are as shown in the figure.

AB → Isothermal expansion **AC** → Adiabatic expansion

Which of the following options is not correct?



(Odisha NEET 2019)

Options:

A. $\Delta S_{\text{isothermal}} > \Delta S_{\text{adiabatic}}$

B. $T_A = T_B$

C. $W_{\text{isothermal}} > W_{\text{adiabatic}}$

D. $T_C > T_A$

Answer: D

Solution:

Solution:

In adiabatic expansion cooling effect will take place, T_C will be less than T_A .

in adiabatic expansion $q = 0$

$$\Delta U = w$$

$$W_{\text{pv}} < 0$$

$$\Delta U < 0$$

$$nC_{\text{vm}} \Delta T < 0$$

$$\Delta T < 0$$

$$T_C - T_A < 0$$

Question22

The bond dissociation energies of X_2 , Y_2 and XY are in the ratio of $1 : 0.5 : 1$. ΔH for the formation of XY is -200 kJ mol^{-1} . The bond dissociation energy of X_2 will be

(NEET 2018)

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Options:

A. 200kJ mol^{-1}

B. 100kJ mol^{-1}

C. 800kJ mol^{-1}

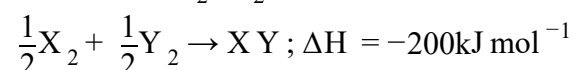
D. 400kJ mol^{-1}

Answer: C

Solution:

Solution:

Let B.E. of X_2 , Y_2 and XY are $x\text{ kJ mol}^{-1}$, $0.5x\text{ kJ mol}^{-1}$ and $x\text{ kJ mol}^{-1}$ respectively.



$$\Delta H = \Sigma(\text{B.E.})_{\text{Reactant}} - \Sigma(\text{B.E.})_{\text{Product}}$$

$$\therefore -200 = \left[\frac{1}{2} \times (x) + \frac{1}{2} \times (0.5x) \right] - [1 \times (x)]$$

$$\text{B.E. of } X_2 = x = 800\text{kJ mol}^{-1}$$

Question23

For a given reaction, $\Delta H = 35.5\text{ kJ mol}^{-1}$ and $\Delta S = 83.6\text{J K}^{-1}\text{mol}^{-1}$. The reaction is spontaneous at (Assume that ΔH and ΔS do not vary with temperature.)

(NEET 2017)

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Options:

A. $T > 425\text{K}$

B. all temperatures

C. $T > 298\text{K}$

D. $T < 425\text{K}$

Answer: A

Solution:

Solution:

For a spontaneous reaction,

$$\Delta G < 0 \text{ i.e., } \Delta H - T \Delta S < 0$$

$$T > \frac{\Delta H}{\Delta S}$$

$$T > \left(\frac{35.5 \times 1000}{83.6} = 424.6 \approx 425\text{K} \right)$$

$$\therefore T > 425\text{K}$$

Question24

A gas is allowed to expand in a well insulated container against a constant external pressure of 2.5 atm from an initial volume of 2.50L to a final volume of 4.50L.

The change in internal energy ΔU of the gas in joules will be (NEET 2017)

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Options:

A. -500J

B. -505J

C. $+505\text{J}$

D. 1136.25J

Answer: B

Solution:

Solution:

$$w = -P_{\text{ext}} \Delta V = -2.5(4.50 - 2.50)$$

$$= -5\text{L atm} = -5 \times 101.325\text{J} = -506.625\text{J}$$

$$\Delta U = q + w$$

As, the container is insulated, thus $q = 0$

$$\text{Hence, } \Delta U = w = -506.625\text{J}$$

Question25

For a sample of perfect gas when its pressure is changed isothermally from p_i to p_f , the entropy change is given by (NEET-II 2016)

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Options:

A. $\Delta S = nR \ln \left(\frac{p_f}{p_i} \right)$

B. $\Delta S = nR \ln \left(\frac{p_i}{p_f} \right)$

C. $\Delta S = nRT \ln \left(\frac{p_f}{p_i} \right)$

D. $\Delta S = RT \ln \left(\frac{p_i}{p_f} \right)$

Answer: B

Solution:

Solution:

For an ideal gas undergoing reversible expansion, when temperature changes from T_i to T_f and pressure changes from p_i to p_f ,

$$\Delta S = nC_p \ln \frac{T_f}{T_i} + nR \ln \frac{p_i}{p_f}$$

For an isothermal process, $T_i = T_f$ so, $\ln 1 = 0$

$$\therefore \Delta S = nR \ln \frac{p_i}{p_f}$$

Question26

The correct thermodynamic conditions for the spontaneous reaction at all temperatures is (NEET- I 2016)

Options:

A. $\Delta H < 0$ and $\Delta S > 0$

B. $\Delta H < 0$ and $\Delta S < 0$

C. $\Delta H < 0$ and $\Delta S = 0$

D. $\Delta H > 0$ and $\Delta S < 0$

Answer: C

Solution:

Solution:

$$\Delta G = \Delta H - T \Delta S$$

If $\Delta H < 0$ and $\Delta S > 0$

$$\Delta G = (-ve) - T (+ve)$$

then at all temperatures, $\Delta G = -ve$, spontaneous reaction

If $\Delta H < 0$ and $\Delta S = 0$

$$\Delta G = (-ve) - T (0) = -ve \text{ at all temperatures.}$$

Question27

**Consider the following liquid-vapour equilibrium. Liquid $\rightleftharpoons$ Vapour
Which of the following relations is correct?
(NEET- I 2016)**

Options:

A. $\frac{d \ln P}{dT} = \frac{-\Delta H_v}{T^2}$

B. $\frac{d \ln P}{dT} = \frac{\Delta H_v}{RT^2}$

C. $\frac{d \ln G}{dT} = \frac{\Delta H_v}{RT^2}$

D. $\frac{d \ln P}{dT} = \frac{-\Delta H_v}{RT}$

Answer: B

Solution:

Solution:

This is Clausius-Clapeyron equation.

Question28

**The heat of combustion of carbon to CO₂ is −393.5kJ/mol. The heat released upon formation of 35.2 g of CO₂ from carbon and oxygen gas is
(2015)**

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Options:

A. +315kJ

B. −630kJ

C. −3.15kJ

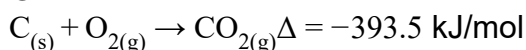
D. −315kJ

Answer: D

Solution:

Solution:

Given :



Amount of heat released on formation of 44 g CO₂ = −393.5kJ

∴ Amount of heat released on formation of 35.2 g CO₂

$$= -\frac{393.5}{44} \times 35.2$$

$$= -314.8 \approx -315\text{kJ}$$

Question29

For the reaction $\text{X}_2\text{O}_4(\text{l}) \rightarrow 2\text{X O}_2(\text{g})$

$\Delta\text{U} = 2.1\text{kcal}$, $\Delta\text{S} = 20\text{cal K}^{-1}$ at 300K

Hence, ΔG is
(2014)

©

Options:

A. 2.7 kcal

B. - 2.7 kcal

C. 9.3 kcal

D. -9.3 kcal

Answer: B

Solution:

Solution:

$$\Delta\text{H} = \Delta\text{U} + \Delta n_{\text{g}} \text{RT}$$

$$\text{Given, } \Delta\text{U} = 2.1\text{kcal}, \Delta n_{\text{g}} = 2$$

$$\text{R} = 2 \times 10^{-3}\text{kcal}, \text{T} = 300\text{K}$$

$$\therefore \Delta\text{H} = 2.1 + 2 \times 2 \times 10^{-3} \times 300 = 3.3\text{kcal}$$

$$\text{Again } \Delta\text{G} = \Delta\text{H} - \text{T } \Delta\text{S}$$

$$\text{Given, } \Delta\text{S} = 20 \times 10^{-3}\text{kcal K}^{-1}$$

On putting the value of ΔH in the equation, we get

$$\Delta\text{G} = 3.3 - 300 \times 20 \times 10^{-3}$$

$$= 3.3 - 6 \times 10^3 \times 10^{-3} = -2.7\text{kcal}$$

Question30

Which of the following statements is correct for the spontaneous adsorption of a gas?
(2014)

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Options:

- A. ΔS is negative and, therefore ΔH should be highly positive.
- B. ΔS is negative and therefore, ΔH should be highly negative.
- C. ΔS is positive and therefore, ΔH should be negative.
- D. ΔS is positive and therefore, ΔH should also be highly positive.

Answer: B

Solution:

Solution:

Using Gibb's -Helmholtz equation,

$$\Delta G = \Delta H - T\Delta S$$

During adsorption of a gas, entropy decreases i.e. $\Delta S < 0$

For spontaneous adsorption, ΔG should be negative, which possible when Δh is highly negative.

Question31

**A reaction having equal energies of activation for forward and reverse reactions has
(2013 NEET)**

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Options:

- A. $\Delta H = 0$
- B. $\Delta H = \Delta G = \Delta S = 0$
- C. $\Delta S = 0$
- D. $\Delta G = 0$

Answer: A

Solution:

Solution:

$$\Delta H = (E_a)_f - (E_a)_b = 0$$

Question32

When 5 litres of a gas mixture of methane and propane is perfectly combusted at 0°C and 1 atmosphere, 16 litres of oxygen at the same temperature and pressure is consumed. The amount of heat released from this combustion in

kJ ($\Delta H_{\text{comb.}}(\text{CH}_4) = 890\text{kJ mol}^{-1}$, $\Delta H_{\text{comb.}}(\text{C}_3\text{H}_8) = 2220\text{kJ mol}^{-1}$)

is
(Karnataka NEET 2013)

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Options:

A. 38

B. 317

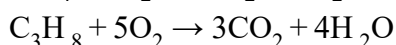
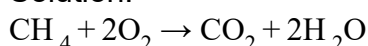
C. 477

D. 32

Answer: B

Solution:

Solution:



$$\text{CH}_4 + \text{C}_3\text{H}_8 = \frac{5}{22.4} = 0.22 \text{ moles}$$

$$\text{O}_2 = \frac{16}{22.4} = 0.71 \text{ moles}$$

$$2x + (0.22 - x)5 = 0.71$$

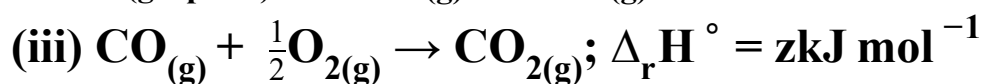
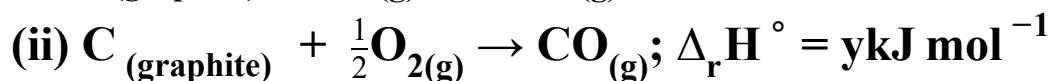
$$x = 0.13$$

Heat liberated

$$= 0.13 \times 890 + 0.09 \times 2220 = 316\text{kJ}$$

Question33

Three thermochemical equations are given below



Based on the above equations, find out which of the relationship given below is correct.

(Karnataka NEET 2013)

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Options:

A. $z = x + y$

B. $x = y + z$

C. $y = 2z - x$

D. $x = y - z$

Answer: B

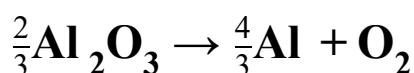
Solution:

Solution:

According to Hess's law, equation (i) is equal to equations (ii) + (iii).

Question34

The Gibb's energy for the decomposition of Al_2O_3 at 500°C is as follows



$$\Delta_{\text{r}}G = +960\text{kJ mol}^{-1}$$

The potential difference needed for the electrolytic reduction of aluminium oxide (Al_2O_3) at 500°C is at least

(2012 Mains)

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Options:

A. 4.5 V

B. 3.0 V

C. 2.5 V

D. 5.0 V

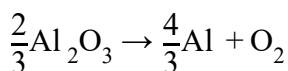
Answer: C

Solution:

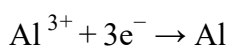
Solution:

$$\Delta G^\circ = -nF E^\circ$$

$$F = 96500, \Delta G^\circ = +960 \times 10^3 \frac{\text{J}}{\text{mol}}$$



$$\text{Total number of Al atoms in Al}_2\text{O}_3 = \frac{2}{3} \times 2 = \frac{4}{3}$$



As 3e^- change occur for each Al atom

$$\therefore \text{Total } n = \frac{4}{3} \times 3 = 4$$

$$E^\circ = \frac{\Delta G^\circ}{nF} = \frac{960 \times 1000}{4 \times 96500}$$

$$\Rightarrow E^\circ = -2.48 = -2.5\text{V}$$

Question35

Equal volumes of two monatomic gases, A and B at same temperature and pressure are mixed. The ratio of specific heats (C_p/C_v) of the mixture will be (2012 Mains)

Options:

A. 0.83

B. 1.50

C. 3.3

D. 1.67

Answer: D

Solution:

Solution:

C_p for monoatomic gas mixture of same volume $= \frac{5}{2}R$

$$C_v = \frac{3}{2}R$$

$$\therefore \frac{C_p}{C_v} = \frac{\frac{5}{2}R}{\frac{3}{2}R} = \frac{5}{3} = 1.67$$

Question36

Standard enthalpy of vaporization $\Delta_{\text{vap}}H^\circ$ for water at 100°C is 40.66kJ mol^{-1} . The internal energy of vaporisation of water at 100°C (in kJ mol^{-1}) is (Assume water vapour to behave like an ideal gas) (2012)

Options:

A. + 37.56

B. - 43.76

C. + 43.76

D. + 40.66

Answer: A

Solution:

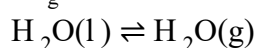
Solution:

$$\Delta_{\text{vap}}H^\circ = 40.66\text{kJ mol}^{-1}$$

$$T = 100 + 273 = 373\text{K}, \Delta E = ?$$

$$\Delta H = \Delta E = \Delta n_g RT \Rightarrow \Delta E = \Delta H - \Delta n_g RT$$

$$\Delta n_g = \text{No. of gaseous mole of products} - \text{No. of gaseous moles of reactants}$$



$$\Delta n_g = 1 - 0 = 1$$

$$\Delta E = \Delta H - RT$$

$$\begin{aligned}\Delta E &= (40.66 \times 10^3) - (8.314 \times 373) \\ &= 37559 \text{ J/mol or } 37.56 \text{ kJ/mol}\end{aligned}$$

Question37

The enthalpy of fusion of water is 1.435 kcal/mol. The molar entropy change for the melting of ice at 0°C is (2012)

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Options:

A. 10.52 cal/(mol K)

B. 21.04 cal/(mol K)

C. 5.260 cal/(mol K)

D. 0.526 cal/(mol K)

Answer: C

Solution:

Solution:

$$\Delta H_f = 1.435 \text{ kcal/mol}$$

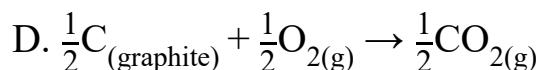
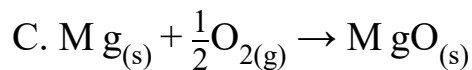
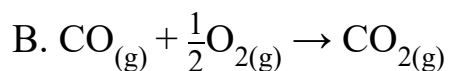
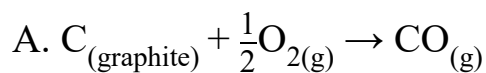
$$\Delta S = \frac{\Delta H_f}{T_f} = \frac{1.435 \times 10^3}{273} = 5.26 \text{ cal/mol K}$$

Question38

In which of the following reactions, standard reaction entropy change (ΔS°) is positive and standard Gibbs energy change (ΔG°) decreases sharply with increasing temperature? (2012)

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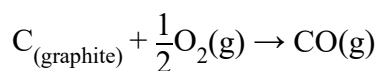
Options:



Answer: A

Solution:

Solution:



$$\Delta n_{\text{g}} = 1 - \frac{1}{2} = \frac{1}{2}$$

As amount of gaseous substance is increasing in product, thus

ΔS is positive for this reaction.

And we know that $\Delta G = \Delta H - T\Delta S$

As ΔS is positive, thus increase in temperature will make $T\Delta S$ more negative and ΔG will decrease

Question39

Consider the following processes:

	$\Delta H(\text{kJ/mol})$
$\frac{1}{2}\text{A} \rightarrow \text{B}$	+ 150
$3\text{B} \rightarrow 2\text{C} + \text{D}$	- 125
$\text{E} + \text{A} \rightarrow 2\text{D}$	+ 350

(2011 Mains)

Options:

A. 525 kJ/mol

B. - 175 kJ/mol

C. -325 kJ/mol

D. 325 kJ/mol

Answer: B

Solution:

Solution:

Adding all the equations :

	ΔH
$A \rightarrow 2B$	300 kJ/mol
$3B \rightarrow 2C + D$	- 125 kJ/mol
$2D \rightarrow A + E$	- 350 kJ/mol
$B + D \rightarrow E + 2C;$	

$$\Delta H = (300 - 125 - 350) = -175 \text{ kJ/mol}$$

Question40

**Which of the following is correct option for free expansion of an ideal gas under adiabatic condition?
(2011)**

©

Options:

A. $q = 0, \Delta T \neq 0, w = 0$

B. $q \neq 0, \Delta T = 0, w = 0$

C. $q = 0, \Delta T = 0, w = 0$

D. $q = 0, \Delta T < 0, w \neq 0$

Answer: C

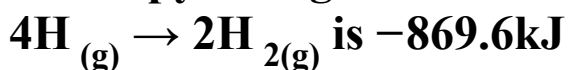
Solution:

Solution:

For free expansion of an ideal gas under adiabatic condition $q = 0$, $\Delta T = 0$, $w = 0$

Question41

Enthalpy change for the reaction,



The dissociation energy of H - H bond is (2011)

©

Options:

A. - 434.8 kJ

B. - 896.6 kJ

C. + 434.8 kJ

D. + 217.4 kJ

Answer: C

Solution:

Solution:

The dissociation energy of H - H bond is $\frac{869.6}{2} = 434.8\text{kJ}$

Question42

If the enthalpy change for the transition of liquid water to steam is 30kJ mol^{-1} at 27°C , the entropy change for the process would be (2011)

©

Options:

A. $10\text{J mol}^{-1}\text{K}^{-1}$

B. $1.0\text{J mol}^{-1}\text{K}^{-1}$

C. $0.1\text{J mol}^{-1}\text{K}^{-1}$

D. $100\text{J mol}^{-1}\text{K}^{-1}$

Answer: D

Solution:

Solution:

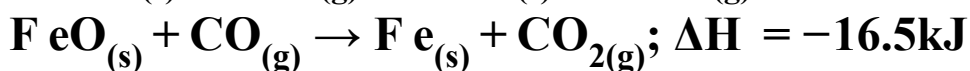
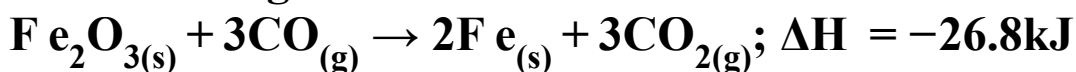
We know that $\Delta G = \Delta H - T\Delta S$

$$0 = \Delta H - T\Delta S \quad [\because \Delta G = 0]$$

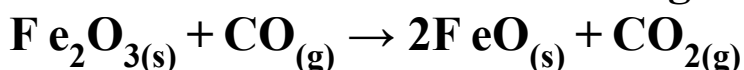
$$\Delta S = \frac{\Delta H}{T} = \frac{30 \times 10^3}{300} = 100\text{J mol}^{-1}\text{K}^{-1}$$

Question43

The following two reactions are known



The value of ΔH for the following reaction



is

(2010 Mains)

Options:

A. $+10.3\text{ kJ}$

B. -43.3 kJ

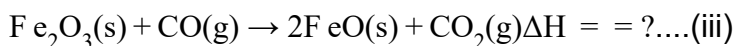
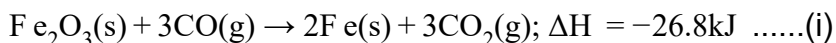
C. -10.3 KJ

D. $+6.2\text{ KJ}$

Answer: D

Solution:

Solution:



Eq (iii) can be obtained as (i) - 2(ii)

$$\Delta H = -26.8 - 2(-16.5) = -26.8 + 33.0 = +6.2\text{kJ}$$

Question44

For vaporization of water at 1 atmospheric pressure, the values of ΔH and ΔS are 40.63K J mol^{-1} and $108.8\text{J K}^{-1}\text{mol}^{-1}$ respectively. The temperature when Gibb's energy change (ΔG) for this transformation will be zero, is (2010 Mains)

©

Options:

A. 273.4 K

B. 393.4 K

C. 373.4 K

D. 293.4 K

Answer: C

Solution:

Solution:

According to Gibb's equation, $\Delta G = \Delta H - T\Delta S$

When $\Delta G=0, \Delta H = T\Delta S$

Given,

$$\Delta H = 40.63\text{kJ mol}^{-1} = 40.63 \times 10^3\text{J mol}^{-1}$$

$$\Delta S = 108.8\text{J K}^{-1}\text{mol}^{-1}$$

$$\therefore T = \frac{\Delta H}{\Delta S} = \frac{40.63 \times 10^3}{108.8} = 373.43\text{K}$$

Question45

**Three moles of an ideal gas expanded spontaneously into vacuum.
The work done will be
(2010 Mains)**

©

Options:

- A. Infinite
- B. 3 Joules
- C. 9 Joules
- D. Zero

Answer: D

Solution:

Solution:

Since the ideal gas expands spontaneously into vacuum,

$P_{\text{ext}} = 0$, hence work done is also zero.

Question46

Match List I (Equations) with List II (Type of processes) and select the correct option.

List (Equation)	List II (types of processes)
(A) $K_p > Q$	(i) Non-spontaneous
(B) $\Delta G^\circ < RT \ln Q$	(ii) Equilibrium
(C) $K_p = Q$	(iii) Spontaneous and endothermic
(D) $T > \frac{\Delta H}{\Delta S}$	(iv) Spontaneous

(2010 Mains)

Options:

A. A - (i), B - (ii), C - (iii), D - (iv)

B. A - (iii), B - (iv), C - (ii), D - (i)

C. A - (iv), B - (i), C - (ii), D - (iii)

D. A - (ii), B - (i), C - (iv), D - (iii)

Answer: C

Solution:

Solution:

When $K_p > Q$, rate of forward reaction > rate of backward reaction.

∴ Reaction is spontaneous

When $\Delta G^\circ < RT \ln Q$, ΔG° is positive, reverse reaction is feasible, thus reaction is non-spontaneous

When $K_p = Q$ rate of forward reaction = rate of backward reaction.

∴ Reaction is equilibrium

When $T \Delta S > \Delta H$, ΔG will be negative only when $\Delta H = +ve$

∴ Reaction is spontaneous and endothermic.

Question47

Standard entropies of X_2 , Y_2 and XY_3 are 60, 40 and 50 $J K^{-1} mol^{-1}$ respectively. For the reaction $\frac{1}{2}X_2 + \frac{3}{2}Y_2 \rightleftharpoons XY_3$, $\Delta H = -30 kJ$, to be at equilibrium the temperature should be (2010)

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Options:

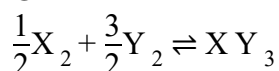
- A. 750 K
- B. 1000 K
- C. 1250 K
- D. 500 K

Answer: A

Solution:

Solution:

Given reaction is :



We know, $\Delta S^\circ = \sum S^\circ_{\text{products}} - \sum S^\circ_{\text{reactants}}$
 $= 50 - (30 + 60) = -40 J K^{-1} mol^{-1}$

At equilibrium $\Delta G^\circ = 0$

$$\Delta H^\circ = T \Delta S^\circ$$

$$\therefore T = \frac{\Delta H^\circ}{\Delta S^\circ} = \frac{-30 \times 10^3 J mol^{-1}}{-40 J K^{-1} mol^{-1}} = 750 K$$

Question 48

For an endothermic reaction, energy of activation is E_a and enthalpy of reactions is ΔH (both of these in kJ/mol). Minimum value of E_a will be (2010)

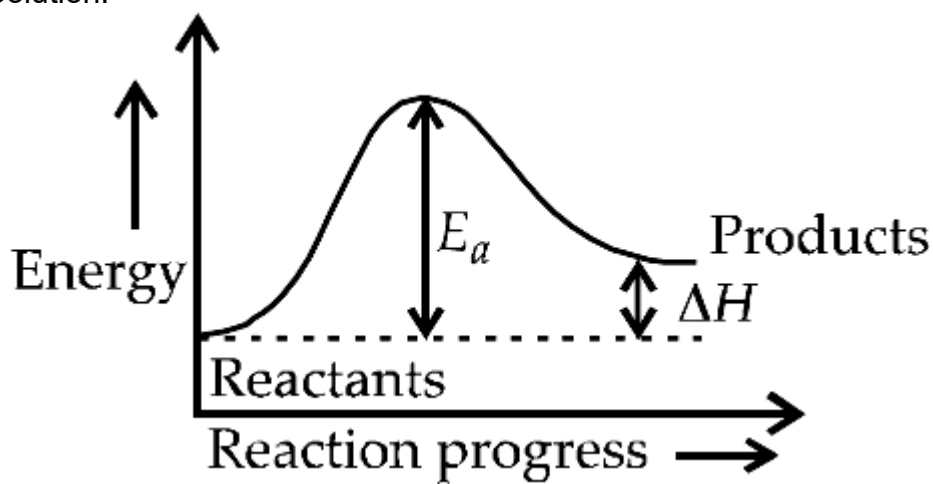
Options:

- A. less than ΔH
- B. equal to ΔH
- C. more than ΔH
- D. equal to zero

Answer: C

Solution:

Solution:



We find that the least E_a will be more than ΔH for an endothermic reaction since $E_{\text{products}} > E_{\text{reactants}}$

Question49

From the following bond energies:

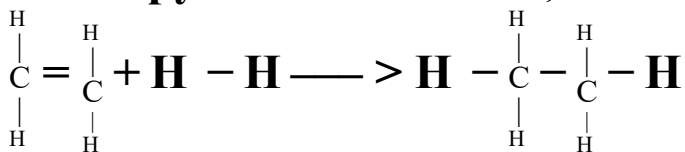
H - H bond energy : $431.37 \text{ kJ mol}^{-1}$

C = C bond energy : $606.10 \text{ kJ mol}^{-1}$

C -C bond energy : $336.9 \text{ kJ mol}^{-1}$

C-H bond energy : $410.50 \text{ kJ mol}^{-1}$

Enthalpy for the reactions,



**will be
(2009)**

Options:

A. $-243.6 \text{ kJ mol}^{-1}$

B. $-120.0 \text{ kJ mol}^{-1}$

C. $553.0 \text{ kJ mol}^{-1}$

D. $1523.6 \text{ kJ mol}^{-1}$

Answer: B

Solution:

Solution:

For the given reaction, enthalpy of reaction can be calculated as

$= \text{B.E.}(\text{reactant}) - \text{B.E.}(\text{product})$

$$= [\text{B.E.}_{(\text{C}=\text{C})} + \text{B.E.}_{(\text{H}-\text{H})} + 4 \times \text{B.E.}_{(\text{C}-\text{H})}] - [\text{B.E.}_{(\text{C}-\text{C})} + 6 \times \text{B.E.}_{(\text{C}-\text{H})}]$$

$$= [606.10 + 431.37 + 4 \times 410.50] - [336.49 + 6 \times 410.50]$$

$$= 2679.47 - 2799.49 = -120.02 \text{ kJ mol}^{-1}$$

Question 50

The values of ΔH and ΔS for the reaction,



are 170 kJ and 170 J K^{-1} , respectively. This reaction will be spontaneous at
(2009)

Options:

A. 910 K

B. 1110 K

C. 510 K

D. 710 K

Answer: B

Solution:

Solution:

For the reaction to be spontaneous,

$$\Delta G = -ve$$

$$\text{Given, } \Delta H = 170\text{kJ} = 170 \times 10^3\text{J}$$

$$\Delta S = 170\text{J K}^{-1}$$

Applying, $\Delta G = \Delta H - T \Delta S$, the value of $\Delta G = -ve$ only when $T \Delta S > \Delta H$, which is possible only when $T = 1110\text{K}$.

$$\therefore \Delta G = 170 \times 10^3 - (1110 \times 170) = -18700\text{J}$$

Thus, reaction is spontaneous at $T = 1110\text{K}$

Question51

Which of the following are not state functions?

(I) $q + w$

(II) q

(III) w

(IV) $H - Ts$

(2008)

Options:

A. (I) (II) and (III)

B. (II) and (III)

C. (I) and (IV)

D. (II) (III) and (IV)

Answer: B

Solution:

Solution:

State functions or state variables are those which depend only on the state of the system and not on how the state was reached.

$$\left. \begin{array}{l} q + w = \Delta E \text{ (internal energy)} \\ H - TS = G \text{ (free energy)} \end{array} \right\} \text{State functions}$$

Path function depends on the path followed during a process as well as the end states. Work and heat are the path functions.

Question52

For the gas phase reaction



Which of the following conditions are correct
(2008)

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Options:

A. $\Delta H < 0$ and $\Delta S < 0$

B. $\Delta H > 0$ and $\Delta S < 0$

C. $\Delta H = 0$ and $\Delta S < 0$

D. $\Delta H > 0$ and $\Delta S > 0$

Answer: D

Solution:

Solution:

Gas phase reaction



$$\Delta H = \Delta E + \Delta n_g RT$$

Δn_g = Change in number of moles of product and reactant species.

Since $\Delta n_g = +ve$, hence $\Delta H = +ve$

also one mole of PCl_5 is dissociated into two moles of PCl_3 and Cl_2 in the same phase.

Therefore, $\Delta S = S_{\text{product}} - S_{\text{reactant}}$

$\Delta S = +ve$.

Question53

Bond dissociation enthalpy of H_2 , Cl_2 and HCl are 434, 242 and 431 kJ mol^{-1} respectively. Enthalpy of formation of HCl is
(2008)

Options:

A. -93kJ mol^{-1}

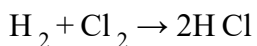
B. 245kJ mol^{-1}

C. 93kJ mol^{-1}

D. -245kJ mol^{-1}

Answer: A**Solution:**

Solution:



$$\begin{aligned}\Delta H_{\text{reaction}} &= \Sigma(\text{B.E})_{\text{reactant}} - \Sigma(\text{B.E})_{\text{product}} \\ &= [(\text{B} \cdot \text{E})_{\text{H-H}} + (\text{B} \cdot \text{E})_{\text{Cl-Cl}}] - [2\text{B} \cdot \text{E}_{(\text{H-Cl})}] \\ &= 434 + 242 - (431) \times 2\end{aligned}$$

$$\Delta H_{\text{reaction}} = -186\text{kJ}$$

Heat of formation is the amount of heat absorbed or evolved when one mole of substance is directly obtained from its constituent element.

Hence, enthalpy of formation of HCl = $\frac{-186}{2}\text{kJ}$

$$= -93\text{kJ mol}^{-1}$$

Question54

Given that bond energies of H - H and Cl - Cl are 430kJ mol^{-1} and 240kJ mol^{-1} respectively and ΔH_f for HCl is -90kJ mol^{-1} bond enthalpy of HCl is (2007)

Options:

A. 380kJ mol^{-1}

B. 425kJ mol^{-1}

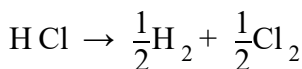
C. 245kJ mol^{-1}

D. 290kJ mol^{-1}

Answer: B

Solution:

Solution:



$$\Delta H = \sum B.E. (\text{products}) - \sum B.E. (\text{reactants})$$

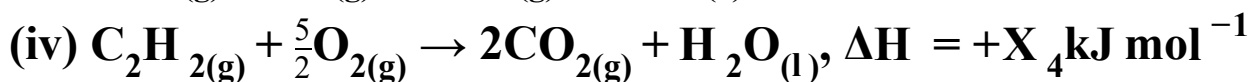
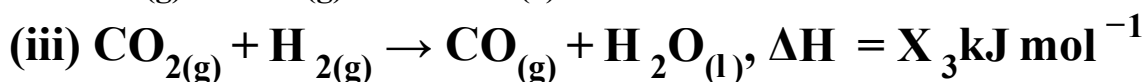
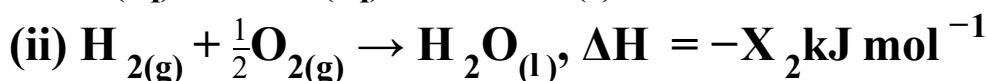
$$= \frac{1}{2} [B.E. (\text{H}_2) + B.E. (\text{Cl}_2)] - B.E. (\text{HCl})$$

$$= \frac{1}{2} (430 + 240) - (-90) = \frac{1}{2} \times 670 + 90$$

$$= 335 + 90 = 425\text{kJ mol}^{-1}$$

Question 55

Consider the following reactions:



Enthalpy of formation of $\text{H}_2\text{O}_{(\text{l})}$ is

(2007)

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Options:

A. $+X_3\text{kJ mol}^{-1}$

B. $-X_4\text{kJ mol}^{-1}$

C. $+X_1\text{kJ mol}^{-1}$

D. $-X_2\text{kJ mol}^{-1}$

Answer: D

Solution:

Solution:

The amount of heat absorbed or released when 1 mole of a substance is directly obtained from its constituent elements is called the heat of formation or enthalpy of formation. Equation (i) represents neutralisation reaction, (iii) represents hydrogenation reaction and (iv) represents combustion reaction.

Question56

The enthalpy of hydrogenation of cyclohexene is -119.5kJ mol^{-1} , If resonance energy of benzene is -150.4kJ mol^{-1} , its enthalpy of hydrogenation would be (2006)

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Options:

A. -358.5kJ mol^{-1}

B. -508kJ mol^{-1}

C. -208.1kJ mol^{-1}

D. -269.9kJ mol^{-1}

Answer: C

Solution:

Solution:

$$\Delta H = -119.5\text{kJ mol}^{-1}$$

Enthalpy of hydrogenation of benzene

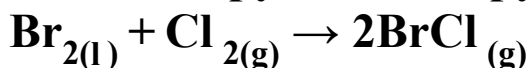
$$= 3 \times \Delta H - \text{resonance energy}$$

$$= 3 \times (-119.5) - (-150.4) = -358.5 + 150.4$$

$$= -208.1\text{kJ mol}^{-1}$$

Question57

The enthalpy and entropy change for the reaction:



are 30 kJ mol^{-1} and $105 \text{ J K}^{-1} \text{ mol}^{-1}$ respectively. The temperature at which the reaction will be in equilibrium is (2006)

©

Options:

A. 300K

B. 285.7 K

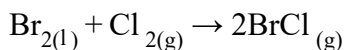
C. 273 K

D. 450 K

Answer: B

Solution:

Solution:



$$\Delta H = 30 \text{ kJ mol}^{-1}, \Delta S = 105 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$\Delta S = \frac{\Delta H}{T} \text{ i.e. } 105 = \frac{30}{T} \times 1000$$

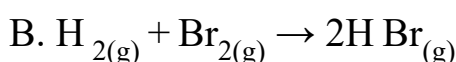
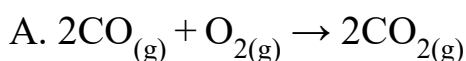
$$\therefore T = \frac{30 \times 1000}{105} = 285.7 \text{ K}$$

Question58

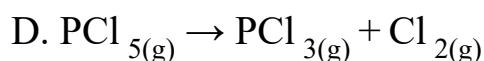
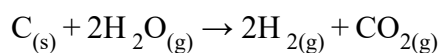
Assume each reaction is carried out in an open container. For which reaction will $\Delta H = \Delta E$? (2006)

©

Options:



C.

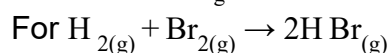


Answer: B

Solution:

Solution:

$$\Delta H = \Delta E + \Delta n_g RT$$



$$\Delta n_g = 2 - (1 + 1) = 0. \text{ i.e. } \Delta H = \Delta E$$

Question 59

Identify the correct statement for change of Gibbs energy for a system (ΔG_{system}) at constant temperature and pressure.

(2006)

Options:

A. If $\Delta G_{\text{system}} < 0$, the process is not spontaneous.

B. If $\Delta G_{\text{system}} > 0$, the process is spontaneous.

C. If $\Delta G_{\text{system}} = 0$, the system has attained equilibrium.

D. If $\Delta G_{\text{system}} = 0$, the system is still moving in a particular direction.

Answer: C

Solution:

Solution:

The criteria for spontaneity of a process in terms of ΔG is as follows:

→ If ΔG is negative, the process is spontaneous.

→ If ΔG is positive, the process does not occur in the forward direction. It may occur in the backward direction.

→ If ΔG is zero, the system is in equilibrium.

Question60

Which of the following pairs of a chemical reaction is certain to result in a spontaneous reaction?
(2005)

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Options:

- A. Exothermic and increasing disorder
- B. Exothermic and decreasing disorder
- C. Endothermic and increasing disorder
- D. Endothermic and decreasing disorder

Answer: A

Solution:

Solution:

For spontaneous reaction $\Delta H = -ve$, $\Delta S = +ve$

Spontaneity depends upon both critical minimum energy and maximum randomness / disorder.

Question61

A reaction occurs spontaneously if
(2005)

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Options:

- A. $T \Delta S < \Delta H$ and both ΔH and ΔS are + ve
- B. $T \Delta S > \Delta H$ and ΔH is +ve and ΔS is -ve
- C. $T \Delta S > \Delta H$ and both ΔH and ΔS are + ve
- D. $T \Delta S = \Delta H$ and both ΔH and ΔS are + ve

Answer: C

Solution:

Solution:

$$\Delta G = \Delta H - T \Delta S$$

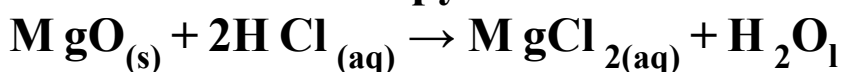
$\Delta G = -ve$ for spontaneous reaction

When $\Delta S = +ve$, $\Delta H = +ve$

and $T \Delta S > \Delta H \Rightarrow \Delta G = -ve$

Question62

The absolute enthalpy of neutralisation of the reaction :



will be

(2005)

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Options:

A. -57.33kJ mol^{-1}

B. greater than -57.33kJ mol^{-1}

C. less than -57.33kJ mol^{-1}

D. 57.33kJ mol^{-1} If

Answer: B

Solution:

Solution:

The enthalpy of hydration of Mg^{2+} ion is very high. Due to this, the absolute enthalpy of neutralization of the reaction is greater than -57.33KJ mol^{-1} . The ΔH for the above reaction is found to be -146KJ which states that the enthalpy of neutralization of H^+ is -73KJ/mol which is greater than the enthalpy of neutralization i.e. 57.33KJ/mol .

Question63

If the bond energies of H – H , Br – Br and H – Br are 433,192 and 364kJ mol⁻¹ respectively, the ΔH° for the reaction $\text{H}_2 + \text{Br}_{2(g)} \rightarrow 2\text{H Br}_{(g)}$ is
(2004)

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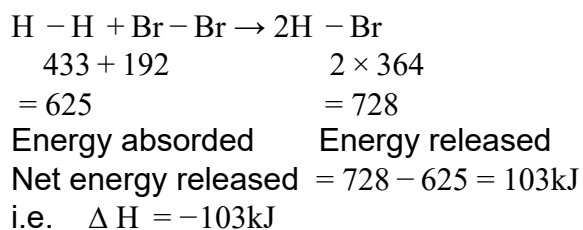
Options:

- A. -261kJ
- B. +103kJ
- C. +261kJ
- D. -103kJ

Answer: D

Solution:

Solution:



Question64

Standard enthalpy and standard entropy changes for the oxidation of ammonia at 298 K are -382.64 and -145.6kJ mol⁻¹, respectively. Standard Gibbs' energy change for the same reaction at 298 K is
(2004)

©

Options:

- A. -221.1kJ mol⁻¹

B. -339.3kJ mol^{-1}

C. -439.3kJ mol^{-1}

D. -523.2kJ mol^{-1}

Answer: B

Solution:

Solution:

$$\Delta G = \Delta H - T \Delta S$$

$$= -382.64 - 298 \left(\frac{-145.6}{1000} \right)$$

$$= -382.64 + 43.38 = -339.3\text{kJ mol}^{-1}$$

Question65

Considering entropy (S) as a thermodynamic parameter, the criterion for the spontaneity of any process is (2004)

©

Options:

A.

$$\Delta S_{\text{system}} + \Delta S_{\text{surroundings}} > 0$$

B.

$$\Delta S_{\text{system}} - \Delta S_{\text{surroundings}} > 0$$

C. $\Delta S_{\text{system}} > 0$ only

D. $\Delta S_{\text{surroundings}} > 0$ only.

Answer: A

Solution:

Solution:

$$\Delta S = R \ln V_2/V_1$$

Here the volume of gas increases from V_1 to V_2 at constant temperature T .

since $V_2 > V_1$, it is obvious that the spontaneous (irreversible) isothermal expansion of a gas is accompanied by an increase in the entropy of the system and its surroundings considered together.

$$\Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$$

Question66

The work done during the expansion of a gas from a volume of 4 dm^3 to 6 dm^3 against a constant external pressure of 3 atm is ($1\text{ Latm} = 101.32\text{ J}$) (2004)

Options:

- A. -6 J
- B. -608 J
- C. $+304\text{ J}$
- D. -304 J

Answer: B

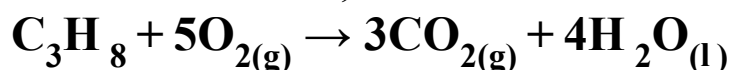
Solution:

Solution:

$$\begin{aligned} \text{Work} &= -P_{\text{ext}} \times \text{volume change} \\ &= -3 \times 101.32 \times (6 - 4) = 6 \times 101.32 \\ &= -607.92\text{ J} \approx -608\text{ J} \end{aligned}$$

Question67

For the reaction,



at constant temperature, $\Delta H - \Delta E$ is (2003)

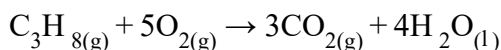
Options:

- A. $+RT$
- B. $-3RT$
- C. $+3RT$
- D. $-RT$

Answer: B

Solution:

Solution:



$$\Delta n_g = 3 - 6 = -3$$

$$\Delta H = \Delta E + P \Delta V \text{ or } \Delta H - \Delta E = P \Delta V$$

$$\Delta H - \Delta E = \Delta n_g RT = -3RT$$

Question68

The densities of graphite and diamond at 298K are 2.25 and 3.31gcm^{-3} , respectively. If the standard free energy difference (ΔG°) is equal to 1895J mol^{-1} , the pressure at which graphite will be transformed into diamond at 298K is (2003)

Options:

- A. $9.92 \times 10^8 \text{Pa}$
- B. $9.92 \times 10^7 \text{Pa}$
- C. $9.92 \times 10^6 \text{Pa}$
- D. $9.92 \times 10^5 \text{Pa}$

E. None of the above

Answer: E

Solution:

Solution:

$$\Delta G = -P \Delta V = \text{Work done}$$

$$\Delta V = \left(\frac{12}{3.31} - \frac{12}{2.25} \right) \times 10^{-3} \text{L} = -1.71 \times 10^{-3} \text{L}$$

$$\Delta G = \text{Work done} = -(-1.71 \times 10^{-3}) \times P \times 101.3 \text{J}$$

$$P = \frac{1895}{1.71 \times 10^{-3} \times 101.3} = 10.93 \times 10^3 \text{atm}$$

$$= 11.07 \times 10^8 \text{Pa}$$

Question69

What is the entropy change (in $\text{J K}^{-1} \text{mol}^{-1}$) when one mole of ice is converted into water at 0°C ? (The enthalpy change for the conversion of ice to liquid water is 6.0kJ mol^{-1} at 0°C .) (2003)

©

Options:

A. 20.13

B. 2.013

C. 2.198

D. 21.98

Answer: D

Solution:

Solution:

$$S = \frac{q_{\text{rev}}}{T} = \frac{6000}{273} = 21.978 \text{J K}^{-1} \text{mol}^{-1}$$

Question70

Formation of a solution from two components can be considered as

(i) Pure solvent $\rightarrow$ separated solvent molecules, ΔH_1

(ii) Pure solute $\rightarrow$ separated solute molecules, ΔH_2

(iii) Separated solvent and solute molecules $\rightarrow$ solution, ΔH_3

Solution so formed will be ideal if

(2003)

©

Options:

A.

$$\Delta H_{\text{soln}} = \Delta H_1 + \Delta H_2 + \Delta H_3$$

B.

$$\Delta H_{\text{soln}} = \Delta H_1 + \Delta H_2 - \Delta H_3$$

C.

$$\Delta H_{\text{soln}} = \Delta H_1 - \Delta H_2 - \Delta H_3$$

D.

$$\Delta H_{\text{soln}} = \Delta H_3 - \Delta H_1 - \Delta H_2$$

Answer: A

Solution:

Solution:

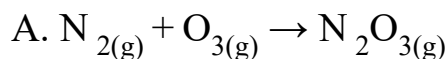
Heat of solution is defined as the amount of heat evolved or absorbed when one mole of the substance is dissolved in excess of the solvent. For hydrated salt and for salts which do not form hydrates, ΔH is positive and for anhydrous salts, it is negative.

Question71

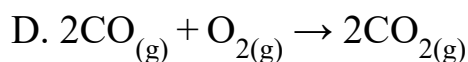
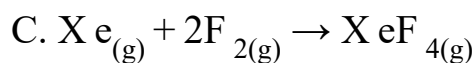
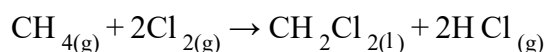
For which one of the following equations is $\Delta H_{\text{reaction}}^{\circ}$ equal to $\Delta H_{\text{f}}^{\circ}$ for the product?
(2003)

©

Options:



B.



Answer: C

Solution:

Solution:

For (c) $\Delta H_{\text{reaction}}^{\circ}$

$$= \Delta H_{\text{f}}^{\circ}(\text{XeF}_4) - [\Delta H_{\text{f}}^{\circ}(\text{Xe}) + 2 \Delta H_{\text{f}}^{\circ}(\text{F}_2)]$$

Enthalpies of formation of elementary substances Xe and F_2 , are taken as zero.

$$\text{Thus, } \Delta H_{\text{reaction}}^{\circ} = \Delta H_{\text{f}}^{\circ}(\text{XeF}_4)$$

Question 72

The molar heat capacity of water at constant pressure, C_p , is $75 \text{ J K}^{-1} \text{ mol}^{-1}$. When 1.0 kJ of heat is supplied to 100 g of water which is free to expand, the increase in temperature of water is
(2003)

©

Options:

A. 1.2 K

B. 2.4K

C. 4.8K

D. 6.6K

Answer: B

Solution:

Solution:

$$\text{Molar heat capacity} = 75 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$18 \text{ g of water} = 1 \text{ mol} \Rightarrow 75 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$1 \text{ g of water} = \frac{75}{18} \text{ J K}^{-1}$$

$$Q = m \cdot C \cdot \Delta t \text{ or } 1000 = 100 \times \frac{75}{18} \times \Delta t$$

$$\Rightarrow \Delta t = \frac{10 \times 18}{75} = 2.4 \text{ K}$$

Question 73

**Unit of entropy is
(2002)**

Options:

A. $\text{J K}^{-1} \text{ mol}^{-1}$

B. J mol^{-1}

C. $\text{J}^{-1} \text{ K}^{-1} \text{ mol}^{-1}$

D. J K mol^{-1}

Answer: A

Solution:

Solution:

Entropy change (ΔS) is given by $\Delta S = q_{\text{rev}}/T$

$\therefore$ Unit of entropy = J/K mol (entropy per unit mole) = $\text{J K}^{-1} \text{ mol}^{-1}$

Question74

In a closed insulated container a liquid is stirred with a paddle to increase the temperature which of the following is true?
(2002)

©

Options:

A. $\Delta E = W \neq 0, q = 0$

B. $\Delta E = W = q \neq 0$

C. $\Delta E = 0, W = q \neq 0$

D. $W = 0, \Delta E = q \neq 0$

Answer: A

Solution:

Solution:

The mathematical form of first law of thermodynamics: $q = \Delta E + W$

since the system is closed and insulated, $q = 0$

Paddle work is done on system. $\therefore W \neq 0$.

Temperature and hence internal energy of the system increases. $\therefore \Delta E \neq 0$

Question75

Heat of combustion ΔH for $C_{(s)}$, $H_{2(g)}$ and $CH_{4(g)}$ are -94 , -68 and -213 kcal/mol then $C_{(s)} + 2H_{2(g)} \rightarrow CH_{4(g)}$ is
(2002)

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Options:

A. -17 kcal

B. -111 kcal

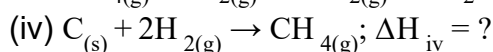
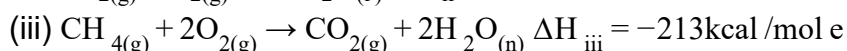
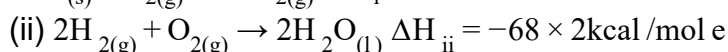
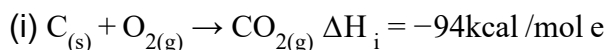
C. -170kcal

D. -85 kcal

Answer: A

Solution:

Solution:



By applying Hess's law we can compute ΔH_{iv}

$$\therefore \Delta H_{\text{iv}} = \Delta H_{\text{i}} + \Delta H_{\text{ii}} - \Delta H_{\text{iii}}$$

$$= (-94 - 68 \times 2 + 213)\text{kcal} = -17\text{kcal}$$

Question76

2 mole of ideal gas at 27°C temperature is expanded reversibly from 2 lit. to 20 lit. Find entropy change. ($R = 2\text{cal/mol K}$) (2002)

Options:

A. 92.1

B. 0

C. 4

D. 9.2

Answer: D

Solution:

Solution:

The change of entropy $dS = \frac{q_{\text{rev}}}{T}$

From the first law of thermodynamics,

$$dq = dU + PdV = C_V dT + PdV$$

$$\Rightarrow \frac{dq}{T} = C_V \frac{dT}{T} + \frac{P}{T} dV$$

$$\Rightarrow \frac{dq}{T} = C_V \frac{dT}{T} + \frac{RdV}{V} \left[\frac{P}{T} = \frac{R}{V} \right]$$

$$\therefore dS = C_V \frac{dT}{T} + R \frac{dV}{V}$$

$$\Rightarrow \Delta S = C_V \ln \frac{T_2}{T_1} + R \ln \frac{V_2}{V_1} \text{ [for one mole of ideal gas]}$$

Here

$$T_2 = T_1 = 27^\circ\text{C} = 300\text{K} \therefore \ln \frac{T_2}{T_1} = 0$$

$$\therefore \Delta S = R \ln \frac{V_2}{V_1} = 2 \ln \frac{20}{2} = 2 \ln 10 = 4.605$$

$$\therefore \Delta S = 4.605 \text{ cal/mol K}$$

Entropy change for 2 moles of gas

$$= 2 \times 4.605 \text{ cal/K} = 9.2 \text{ cal/K}$$

Question 77

Change in enthalpy for reaction, $2\text{H}_2\text{O}_{2(l)} \rightarrow 2\text{H}_2\text{O}_{(l)} + \text{O}_2$ if heat of formation of $\text{H}_2\text{O}_{2(l)}$ and $\text{H}_2\text{O}_{(l)}$ are -188 and -286 kJ/mol respectively, is (2001)

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Options:

A. -196 kJ/mol

B. $+196 \text{ kJ/mol}$

C. $+948 \text{ kJ/mol}$

D. -948 kJ/mol

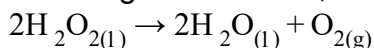
Answer: A

Solution:

Solution:

$$\Delta H_f^\circ = \sum H_{f(\text{product})}^\circ - \sum H_{f(\text{reactant})}^\circ$$

For the given reaction,



$$\begin{aligned}\Delta H_f^\circ &= 2 \times \Delta H_{f(\text{H}_2\text{O})}^\circ - 2 \times \Sigma H_{f(\text{H}_2\text{O}_2)}^\circ \\ &= 2 \times -286 \text{ kJ mol}^{-1} - 2 \times (-188) \text{ kJ mol}^{-1} \\ &= -196 \text{ kJ mol}^{-1}\end{aligned}$$

Question 78

When 1 mol of gas is heated at constant volume temperature is raised from 298 to 308 K. Heat supplied to the gas is 500J Then which statement is correct?
(2001)

©

Options:

- A. $q = w = 500\text{J}$, $\Delta E = 0$
- B. $q = \Delta E = 500\text{J}$, $w = 0$
- C. $q = w = 500\text{J}$, $\Delta E = 0$
- D. $\Delta E = 0$, $q = w = -500\text{J}$

Answer: B

Solution:

Solution:

$$\Delta H = \Delta E + P \Delta V$$

When $\Delta V = 0$; $w = 0$

$$\Delta H = \Delta E + 0 \text{ or } \Delta H = \Delta E$$

As $\Delta E = q + w$, $\Delta E = q$

In the present problem, $\Delta H = 500\text{J}$,

$$\Delta H = \Delta E = 500\text{J}, q = 500\text{J}, w = 0$$

Question 79

Enthalpy of $\text{CH}_4 + 1/2\text{O}_2 \rightarrow \text{CH}_3\text{OH}$ is negative. If enthalpy of combustion of CH_4 and CH_3OH are x and

y respectively. Then which relation is correct?
(2001)

Options:

A. $x > y$

B. $x < y$

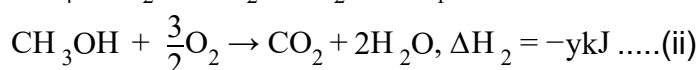
C. $x = y$

D. $x \geq y$

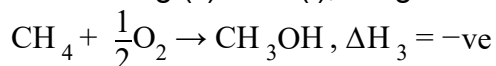
Answer: A

Solution:

Solution:



Subtracting (ii) from (i), we get



i. e., $-x - (-y) = -ve$

$$y - x = -ve$$

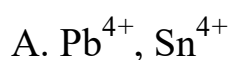
Hence, $x > y$

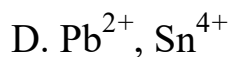
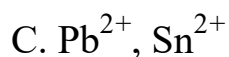
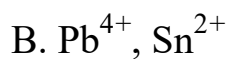
Question80



Most probable oxidation state of Pb and Sn will be
(2001)

Options:





Answer: D

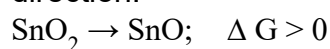
Solution:

Solution:

The sign and magnitude of Gibbs free energy is a criterion of spontaneity for a process. When

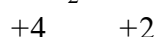
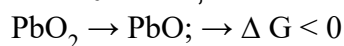
$\Delta G > 0$ or +ve, it means $G_{\text{product}} > G_{\text{reactant}}$

the reaction will not take place spontaneously, i.e. the reaction should be spontaneous in reverse direction.



(more favourable)

$\Delta G < 0$ or -ve, the reaction or change occurs spontaneously.



(more favourable)

Question 81

**Cell reaction is spontaneous when
(2000)**

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Options:

A. ΔG° is negative

B. ΔG° is positive

C. $\Delta E_{\text{red}}^\circ$ is positive

D. $\Delta E_{\text{red}}^\circ$ is negative.

Answer: A

Solution:

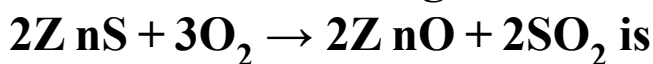
Solution:

For a cell reaction to be spontaneous ΔG° should be negative. As $\Delta G^\circ = -nF E_{\text{cell}}^\circ$ so the value will be -ve only when E_{cell}° is +ve.

Question82



ΔG° for the following reaction



(2000)

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Options:

A. -731 J

B. -1317 J

C. -501 J

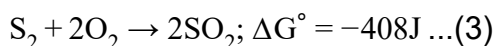
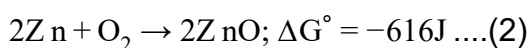
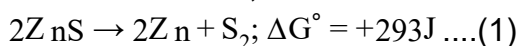
D. +731 J

Answer: A

Solution:

Solution:

For the reactions,



The ΔG° for the reaction,

$2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$ can be obtained by adding eq. (1),(2) and (3)

$$\text{So, } \Delta G^\circ = 293 - 616 - 408 = -731 \text{ J}$$

Question83

At 27° C latent heat of fusion of a compound is 2930J /mol . Entropy change is
(2000)

©

Options:

- A. 9.77J /mol K
- B. 10.77J /mol K
- C. 9.07J /mol K
- D. 0.977J /mol K

Answer: A

Solution:

Solution:

$$\Delta S = \frac{Q}{T} = \frac{2930}{300} = 9.77 \text{ J /mol K}$$

Question84

For the reaction,
 $\text{C}_2\text{H}_5\text{OH}_{(l)} + 3\text{O}_2 \rightarrow 2\text{CO}_{2(g)} + 3\text{H}_2\text{O}_{(l)}$
which one is true?
(2000)

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Options:

- A. $\Delta H = \Delta E - RT$
- B. $\Delta H = \Delta E + RT$
- C. $\Delta H = \Delta E + 2RT$

D. $\Delta H = \Delta E - 2RT$

Answer: A

Solution:

Solution:

$$\Delta H = \Delta E + P \Delta V$$

also $PV = nRT$ (ideal gas equation)

$$\text{or } P \Delta V = \Delta n_g RT$$

Δn_g = change in number of gaseous moles

$$\therefore \Delta H = \Delta E + \Delta n_g RT \Rightarrow \Delta n_g = 2 - 3 = -1$$

$$\Rightarrow \Delta H = \Delta E - RT$$

Question85

In an endothermic reaction, the value of ΔH is (1999)

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Options:

A. negative

B. positive

C. zero

D. constant.

Answer: B

Solution:

Solution:

In endothermic reactions, energy of reactants is less than energy of products. Thus, $E_R < E_P$

$$\Delta H = E_P - E_R = +ve$$

Question86

In the reaction: $S + \frac{3}{2}O_2 \rightarrow SO_3 + 2x\text{kcal}$ and

**$SO_2 + \frac{1}{2}O_2 \rightarrow SO_3 + y\text{kcal}$, the heat of formation of SO_2 is
(1999)**

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Options:

A. $(2x + y)$

B. $(x - y)$

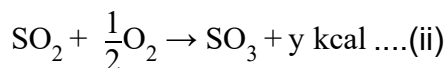
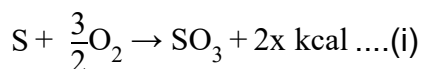
C. $(x + y)$

D. $(2x - y)$

Answer: D

Solution:

Solution:



By subtracting equation (ii) from (i) we get,
The heat of formation of SO_2 is $(2x - y)$ kcal/mole.

Question87

**Identify the correct statement regarding entropy.
(1998)**

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Options:

A. At absolute zero of temperature, the entropy of all crystalline substances is taken to be zero.

B. At absolute zero of temperature, the entropy of a perfectly crystalline substance is +ve.

C. At absolute zero of temperature, entropy of a perfectly crystalline substance is taken to be zero.

D. At 0°C , the entropy of a perfectly crystalline substance is taken to be zero.

Answer: C

Solution:

Solution:

The entropy of a substance increases with increase in temperature. However at absolute zero the entropy of a perfectly crystalline substance is taken as zero, which is also called as third law of thermodynamics

Question88

One mole of an ideal gas at 300K is expanded isothermally from an initial volume of 1 litre to 10 litres. The ΔE for this process is ($R = 2 \text{ cal mol}^{-1}\text{K}^{-1}$) (1998)

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Options:

A. 1381.1cal

B. zero

C. 163.7cal

D. 9L atm

Answer: B

Solution:

Solution:

Change in internal energy depends upon temperature. At constant temperature, the internal energy of the gas remains constant, so $\Delta E = 0$

Question89

Given that $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$, $\Delta H^\circ = -x\text{kJ}$

$2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$, $\Delta H^\circ = -y\text{kJ}$

The enthalpy of formation of carbon monoxide will be
(1997)

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Options:

A. $\frac{y-2x}{2}$

B. $2x - y$

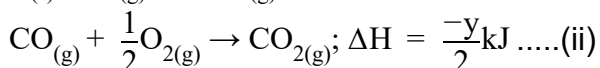
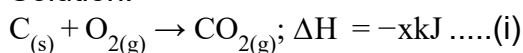
C. $y - 2x$

D. $\frac{2x-y}{2}$

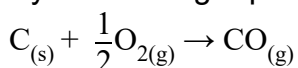
Answer: A

Solution:

Solution:



By subtracting equation (ii) from (i) we get,



$$\Delta H = -x - \left(-\frac{y}{2}\right) = \frac{y-2x}{2}\text{kJ}$$

Question90

Which of the following is the correct equation?
(1996)

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Options:

A. $\Delta U = \Delta W + \Delta Q$

B. $\Delta U = \Delta Q - W$

C. $\Delta W = \Delta U + \Delta Q$

D. None of these

Answer: B

Solution:

Solution:

This is the mathematical relation of first law of thermodynamics. Here ΔU = change in internal energy; ΔQ = heat absorbed by the system and W = work done by the system.

Question91

It enthalpies of formation for C_2H_4 , $CO_{2(g)}$ and $H_2O_{(l)}$ at $25^\circ C$ and 1 atm pressure are 52, -394 and $-286 kJ/mol$ respectively, then enthalpy of combustion of $C_2H_{4(g)}$ will be (1995)

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Options:

A. $+141.2 kJ/mol$

B. $+1412 kJ/mol$

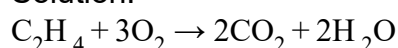
C. $-141.2 kJ/mol$

D. $-1412 kJ/mol$

Answer: D

Solution:

Solution:



$$\Delta H = \Delta H_{\text{products}} - \Delta H_{\text{reactants}}$$

$$= 2 \times (-394) + 2 \times (-286) - (52 + 0) = -1412 kJ/mol$$

Question92

For a reaction to occur spontaneously (1995)

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Options:

- A. ΔH must be negative
- B. ΔS must be negative
- C. $(\Delta H - T \Delta S)$ must be negative
- D. $(\Delta H + T \Delta S)$ must be negative.

Answer: C

Solution:

Solution:

For a reaction to be spontaneous, ΔG (Gibbs' free energy change) must be negative.

$$\Delta G = \Delta H - T \Delta S$$

ΔH = change in enthalpy, ΔS = change in entropy.

Question93

During isothermal expansion of an ideal gas, its (1994,1991)

©

Options:

- A. internal energy increases
- B. enthalpy decreases
- C. enthalpy remains unaffected
- D. enthalpy reduces to zero.

Answer: C

Solution:

Solution:

An isothermal process is a change of a system, in which the temperature remains constant. For an ideal gas during an isothermal expansion the enthalpy, as well as internal energy, remains constant.

During isothermal expansion of an ideal gas, $\Delta E = 0$, $\Delta T = 0$

From the definition of enthalpy, $H = E + PV$

or $\Delta H = \Delta E + \Delta (PV)$

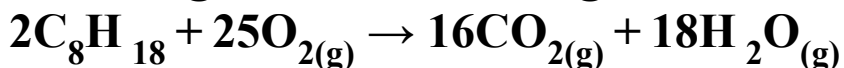
or $\Delta H = \Delta E + \Delta (nRT)$ \{ Since, $PV = nRT$ for an ideal gas \}

or $\Delta H = \Delta E + nR \Delta T$

or $\Delta H = 0$

Question94

Following reaction occurring in an automobile



The sign of ΔH , ΔS and ΔG would be
(1994)

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Options:

A. -, +, +

B. +, +, -

C. +, -, +

D. -, +, -

Answer: D

Solution:

Solution:

The given reaction is a combustion reaction, therefore ΔH is less than 0. Hence, ΔH is negative.

(ii) Since there is increase in the number of moles, therefore ΔS is positive

(iii) Since reaction is spontaneous, therefore ΔG is negative.

Question95

For the reaction $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$, $\Delta H = ?$
(1991)

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Options:

- A. $\Delta E + 2RT$
- B. $\Delta E - 2RT$
- C. $\Delta H = RT$
- D. $\Delta E - RT$

Answer: B

Solution:

Solution:

$$\Delta n_g = 2 - 4 = -2, \Delta H = \Delta E - 2RT$$

Question96

If ΔH is the change in enthalpy and ΔE , the change in internal energy accompanying a gaseous reaction, then
(1990)

©

Options:

- A. ΔH is always greater than ΔE
- B. $\Delta H < \Delta E$ only if the number of moles of the products is greater than the number of moles of the reactants
- C. ΔH is always less than ΔE
- D. $\Delta H < \Delta E$ only if the number of moles of products is less than the number of moles of the reactants.

Answer: D

Solution:

Solution:

If $n_p < n_r$; $\Delta n_g = n_p - n_r = -ve$.

Hence $\Delta H < \Delta E$.
